

Calculus III — Integration Notation Reference

Line integrals · Surface integrals · Vector fields · Green, Stokes & Divergence theorems

1 • The Dimensional Ladder START HERE

Every integral in this course lives at exactly one rung of a dimensional ladder. The **subscript** on $\int$ tells you which rung. The **differential** tells you what kind of infinitesimal chunk you're summing.

0-DIMENSIONAL 0 Points → Endpoints a, b → Boundary of a curve → $f(b) - f(a)$	1-DIMENSIONAL 1 Curves C → Subscript: C → Scalar: ds → Vector: dr → Param: dt	2-DIMENSIONAL 2 Regions R/D · Surfaces S → Subscripts: R, D (flat), S (curved) → Flat area: $dA = dx$ dy → Surface scalar: dS → Surface vector: $d\mathbf{S}$ (bold) → Param: $du\ dv$	3-DIMENSIONAL 3 Volumes V/E → Subscripts: V, E, W → $dV = dx\ dy\ dz$ → Spherical/cylindrical coords
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⚠ The single most confusing notation clash

dS (lowercase italic) = scalar area element (a positive number, like ds for arc length)

$d\mathbf{S}$ or $d\vec{S}$ (bold / vector) = vector surface element = $\mathbf{n}\ dS$ (a vector pointing outward)

Context and bold-font are your only clues. When in doubt: if it's inside $\mathbf{F} \cdot d\mathbf{S}$, it's the vector one.

2 · Notation Decoder

Subscripts — what you're integrating over

SYMBOL	DIMENSION	WHAT IT IS	TYPICAL SETTING
C	1D	Curve (path) in $\mathbb{R}^2$ or $\mathbb{R}^3$	Line integrals
∂C	0D	Boundary of C = its two endpoints	FTC for line integrals
R or D	2D	Flat region in $\mathbb{R}^2$ (a "domain")	Double integrals, Green's Thm
∂D or ∂R	1D	Boundary of region = a closed curve	Green's Theorem LHS
S	2D	Surface in $\mathbb{R}^3$ (a 2D manifold)	Surface integrals, Stokes/Div
∂S	1D	Boundary of surface S = a closed curve	Stokes' Theorem LHS
V or E	3D	Volume (solid region) in $\mathbb{R}^3$	Triple integrals, Div Thm
∂V or ∂E	2D	Boundary of volume = a closed surface	Divergence Theorem LHS

Differentials — how you're measuring infinitesimal pieces

SYMBOL	TYPE	WHAT IT MEASURES	HOW COMPUTED
dt	param	Infinitesimal step in parameter space	Bare parameter; needs $ \mathbf{r}'(t) $ to become geometric
ds	scalar	Arc length element (always positive)	$ds = \mathbf{r}'(t) dt = \sqrt{(x'^2+y'^2+z'^2)} dt$
$d\mathbf{r}$ or $d\mathbf{r}$	vector	Directed displacement along curve	$d\mathbf{r} = \mathbf{r}'(t) dt = (dx, dy, dz)$; note $d\mathbf{r} = \mathbf{T} ds$
$dx dy$ or dA	scalar	Area element in flat region	Cartesian; also $r dr d\theta$ (polar), or $ J du dv$
$du dv$	param	Parameter-space area element	Bare; needs $ \mathbf{r}_u \times \mathbf{r}_v $ to become geometric
dS (scalar)	scalar	Surface area element (always positive)	$dS = \mathbf{r}_u \times \mathbf{r}_v du dv$
$d\mathbf{S}$ (vector)	vector	Oriented surface element (normal vector $\times$ area)	$d\mathbf{S} = (\mathbf{r}_u \times \mathbf{r}_v) du dv = \mathbf{n} dS$
dV	scalar	Volume element	$dx dy dz$; or $\rho^2 \sin \phi d\rho d\phi d\theta$; or $r dr d\theta dz$

Key insight: $\mathbf{T}$ and $\mathbf{n}$ are unit vectors built into the differentials

$d\mathbf{r} = \mathbf{T} ds$ — the tangent unit vector times arc length

$d\mathbf{S} = \mathbf{n} dS$ — the unit normal times surface area

So $\mathbf{F} \cdot d\mathbf{r} = \mathbf{F} \cdot \mathbf{T} ds$ (work = tangential component $\times$ length) and $\mathbf{F} \cdot d\mathbf{S} = \mathbf{F} \cdot \mathbf{n} dS$ (flux = normal component $\times$ area)

3 · Master Table of All Integral Types

Colour of left border = dimension of integration domain.

NAME	NOTATION	PHYSICAL MEANING	PARAMETERISED FORM (HOW YOU COMPUTE IT)	WHEN ZERO / SPECIAL
Scalar line integral	$\int_C f \, ds$	Mass of wire C with density f; or weighted arc length	$\int_a^b f(r(t)) r'(t) \, dt$	$f = 0$ everywhere on C
Vector line integral (work)	$\int_C \mathbf{F} \cdot d\mathbf{r}$	Work done by force $\mathbf{F}$ along C	$\int_a^b \mathbf{F}(r(t)) \cdot r'(t) \, dt$	0 if $\mathbf{F} \perp C$ everywhere; 0 around any closed C if $\text{curl } \mathbf{F} = 0$ (conservative)
Line integral (component form)	$\int_C P \, dx + Q \, dy$	Same as $\mathbf{F} \cdot d\mathbf{r}$ with $\mathbf{F} = (P, Q)$	$\int_a^b [P x'(t) + Q y'(t)] \, dt$	Related to Green's theorem
Line integral in $\mathbb{R}^3$	$\int_C P \, dx + Q \, dy + R \, dz$	$\mathbf{F} \cdot d\mathbf{r}$ with $\mathbf{F} = (P, Q, R)$	$\int_a^b [P x' + Q y' + R z'] \, dt$	0 on closed C if $\text{curl } \mathbf{F} = 0$
Double integral	$\iint_R f \, dA$	Mass of flat plate with density f; volume under surface	$\iint f(x, y) \, dx \, dy$ (iterate, or use polar)	$f=1$ gives area of R
Scalar surface integral	$\iint_S f \, dS$	Mass of shell S with density f	$\iint f(r(u, v)) r_u \times r_v \, du \, dv$	$f=1$ gives surface area of S
Flux integral (vector surface)	$\iint_S \mathbf{F} \cdot d\mathbf{S}$	Net fluid flux through S; flow rate	$\iint \mathbf{F}(r(u, v)) \cdot (r_u \times r_v) \, du \, dv$	0 if $\mathbf{F} \perp \mathbf{n}$ everywhere; 0 on closed S if $\text{div } \mathbf{F} = 0$
Flux over $z=g(x, y)$	$\iint_S \mathbf{F} \cdot d\mathbf{S}$	Flux through an explicit surface	$\iint_D (-P g_x - Q g_y + R) \, dA$ (upward normal)	Sign depends on orientation
Triple integral	$\iiint_V f \, dV$	Mass of solid with density f	$\iiint f(x, y, z) \, dx \, dy \, dz$	$f=1$ gives volume of V

Reading the notation instantly

Count the integral signs → that's the dimension.

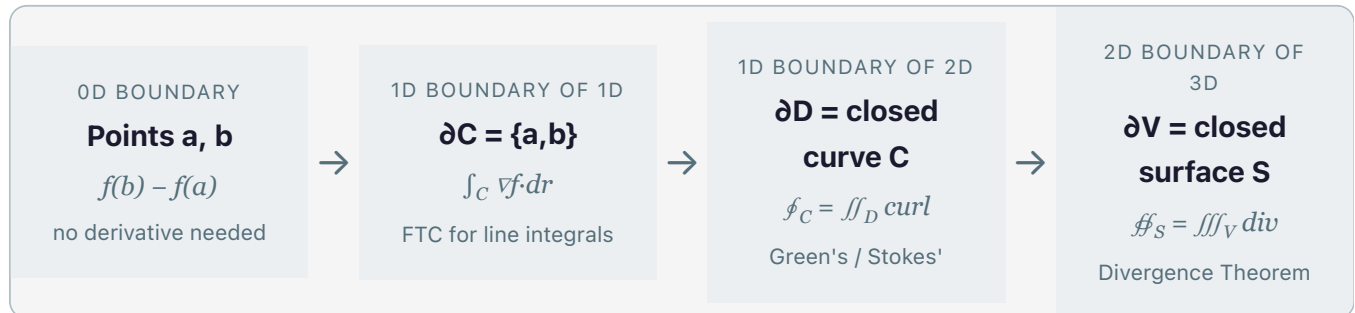
Read the subscript → that's the object.

Read the differential: *lowercase d + lowercase letter* = scalar; *bold d + uppercase* = vector; *dt or du dv* = parameter (needs rescaling).

4 · The Big Theorems UNIFYING THEME

Every major theorem says the same thing in different dimensions:

f over boundary = f of derivative over interior



Fundamental Theorem for Line Integrals

CONSERVATIVE FIELD: $\mathbf{F} = \nabla f$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{b}) - f(\mathbf{a})$$

1D integral → 0D boundary values

Requires: $\mathbf{F} = \nabla f$ (conservative). Path doesn't matter, only endpoints. If C is closed: result = 0.

Green's Theorem

2D REGION D , BOUNDARY $\partial D = C$
(COUNTERCLOCKWISE)

$$\oint_C P dx + Q dy = \iint_D (\partial Q / \partial x - \partial P / \partial y) dA$$

1D line integral = 2D area integral of curl

Requires: D flat (in xy -plane), C simple closed counterclockwise, $\mathbf{F}$ smooth on D .

$\partial Q / \partial x - \partial P / \partial y = (\text{curl } \mathbf{F}) \cdot \mathbf{k}$ = the z -component of curl.

Stokes' Theorem

SURFACE S IN $\mathbb{R}^3$, BOUNDARY $\partial S = C$ (RIGHT-HAND RULE)

$$\oint_C \mathbf{F} \cdot d\mathbf{r} = \iint_S (\text{curl } \mathbf{F}) \cdot d\mathbf{S}$$

1D line integral = 2D surface integral of curl $\mathbf{F}$

Green's theorem is Stokes' with S being a flat region in the xy -plane!

If $\text{curl } \mathbf{F} = 0$ everywhere, then $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for any closed C .

Divergence Theorem (Gauss's)

VOLUME E , CLOSED BOUNDARY SURFACE $\partial E = S$ (OUTWARD NORMAL)

$$\oint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \text{div } \mathbf{F} dV$$

2D flux integral = 3D volume integral of div $\mathbf{F}$

If $\text{div } \mathbf{F} = 0$, total flux out of any closed surface is 0 (incompressible flow).

Use when direct surface integral is hard but volume integral is easy (or vice versa).

The Operator Ladder: gradient $\rightarrow$ curl $\rightarrow$ divergence

The derivative that appears in each theorem follows a fixed pattern:

∇f (gradient of scalar $\rightarrow$ vector) · FTC for line integrals

$\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$ (of vector $\rightarrow$ vector) · Green's & Stokes'

$\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$ (of vector $\rightarrow$ scalar) · Divergence Theorem

And note: $\text{div}(\text{curl } \mathbf{F}) = 0$ always, $\text{curl}(\nabla f) = 0$ always — these are the "boundary of a boundary = 0" identities.

5 · Conservative Vector Fields — Special Cases

What makes a field conservative?

$\mathbf{F} = \nabla f$ for some scalar potential f (equivalently: $\text{curl } \mathbf{F} = 0$ on a simply-connected domain)

Conservative field

- $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends only on endpoints
- $\oint_C \mathbf{F} \cdot d\mathbf{r} = 0$ for any simple closed curve C
- $\text{curl } \mathbf{F} = 0$ (irrotational)
- Find f by integrating: $\partial f / \partial x = P$, $\partial f / \partial y = Q$, $\partial f / \partial z = R$
- Use FTC: $\int_C \mathbf{F} \cdot d\mathbf{r} = f(\text{end}) - f(\text{start})$
- Green/Stokes gives $0 = 0$ (trivially true)

Non-conservative field

- $\int_C \mathbf{F} \cdot d\mathbf{r}$ depends on path
- $\oint_C \mathbf{F} \cdot d\mathbf{r} \neq 0$ in general
- $\text{curl } \mathbf{F} \neq 0$ (has rotation)
- Use Green's/Stokes' to convert hard loop integral to easier area integral
- Closing a path can simplify: split $C = C_1 + (-C_2)$ where C_2 is easy

Simple closed curves in non-conservative fields (using Green's)

SITUATION	APPROACH	RESULT
$\oint_C \mathbf{F} \cdot d\mathbf{r}$, C counterclockwise around D	Green's: convert to $\iint_D (Q_x - P_y) dA$	Area integral, often much easier
$\oint_C \mathbf{F} \cdot d\mathbf{r}$, C clockwise	Reverse orientation: add minus sign	$-\iint_D (Q_x - P_y) dA$
$\text{curl } \mathbf{F} = 0$ but domain has hole (e.g. $\mathbf{F} = (-y, x)/(x^2 + y^2)$)	Domain not simply connected! $\text{curl}=0$ doesn't imply conservative	$\oint = 2\pi \times (\text{winding number around hole})$
Area of region via Green's	$A = \frac{1}{2} \oint_C x dy - y dx$	Clever use: $P = -y/2$, $Q = x/2$, $Q_x - P_y = 1$

6 · Parameterisation — Getting to

dt

or

du dv

You always integrate in parameter space. The *scaling factor* converts parameter area/length to geometric area/length.

Curve parameterisation $\mathbf{r}(t)$, $t \in [a,b]$

INTEGRAL	CONVERTS TO
$\int_C f \, ds$	$\int_a^b f(\mathbf{r}(t)) \, \mathbf{r}' \, dt$
$\int_C \mathbf{F} \cdot d\mathbf{r}$	$\int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$

Trivial reparameterisation

If you replace t with $s=\phi(t)$ (same curve, different speed), the integral is unchanged as long as direction is preserved. Reversing direction flips sign of $\mathbf{F} \cdot d\mathbf{r}$ but not $\int f \, ds$.

Surface parameterisation $\mathbf{r}(u,v)$, $(u,v) \in D$

INTEGRAL	CONVERTS TO
$\iint_S f \, dS$	$\iint_D f(\mathbf{r}(u,v)) \, \mathbf{r}_u \times \mathbf{r}_v \, du \, dv$
$\iint_S \mathbf{F} \cdot d\mathbf{S}$	$\iint_D \mathbf{F}(\mathbf{r}(u,v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) \, du \, dv$

Cross product is your scaling factor

$\mathbf{r}_u \times \mathbf{r}_v$ is perpendicular to the surface and its magnitude = area stretch factor.
For scalar: take $|\mathbf{r}_u \times \mathbf{r}_v|$. For vector flux: keep $(\mathbf{r}_u \times \mathbf{r}_v)$ as a vector — it encodes both area and orientation.

Common non-parameterised / shortcut forms

SURFACE TYPE	SCALAR DS	VECTOR DS (UPWARD)
$z = g(x,y)$ over D	$dS = \sqrt{1 + g_x^2 + g_y^2} \, dA$	$d\mathbf{S} = (-g_x, -g_y, 1) \, dA$
$x = g(y,z)$ over D	$dS = \sqrt{1 + g_y^2 + g_z^2} \, dA$	$d\mathbf{S} = (1, -g_y, -g_z) \, dA$
Sphere $\rho=a$ (spherical coords)	$dS = a^2 \sin \varphi \, d\varphi \, d\theta$	$d\mathbf{S} = a^2 \sin \varphi \, \hat{\mathbf{r}} \, d\varphi \, d\theta$
Cylinder $r=a$ (cylindrical)	$dS = a \, d\theta \, dz$	$d\mathbf{S} = a \, \hat{\mathbf{r}} \, d\theta \, dz$

7 • Mnemonics & Memory Aids

The Wire / Shell / Solid analogy

Scalar integrals = **weighing things**:

- Wire (1D) with density $f \rightarrow \int_C f \, ds$
- Shell (2D surface) with density $f \rightarrow \iint_S f \, dS$
- Solid (3D) with density $f \rightarrow \iiint_V f \, dV$

$f=1$ always gives total length/area/volume.

Flow analogies

Vector integrals = **measuring flow**:

- Work = flow *along* curve $\rightarrow \int_C \mathbf{F} \cdot d\mathbf{r}$ (tangential)
- Flux = flow *through* surface $\rightarrow \iint_S \mathbf{F} \cdot d\mathbf{S}$ (normal)

Work cares about the *tangential* component.

Flux cares about the *normal* component.

Differential sizing rule

Lowercase scalar: always a magnitude, always positive

ds , dS (italic), dA , dV

Bold/vector: has direction, can be negative

$d\mathbf{r}$, $d\mathbf{S}$ (bold)

Parameter: geometric meaning only after multiplying by the scaling factor

dt , $du \, dv$

Boundary orientation memory

Green's: walk CCW around D , region stays on your LEFT.

Stokes': curl your right hand fingers in direction of C ; thumb points in direction of $\mathbf{n}$ on S .

Divergence: normal on ∂E always points *outward* from the volume.

↑ Which theorem to reach for?

Hard **closed curve** integral in 2D $\rightarrow$ **Green's**
(become area integral)

Hard **closed curve** integral in 3D $\rightarrow$ **Stokes'**
(become surface integral)

Hard **closed surface** flux integral $\rightarrow$ **Divergence**
(become volume integral)

$\mathbf{F} = \nabla f?$ $\rightarrow$ **FTC**: just evaluate f at endpoints

The " ∂ " unifier

Every theorem: $\int_{\partial\Omega} = \int_{\Omega} (\text{d-operator})$

- ∂C = two endpoints; d-op = gradient
- ∂D = closed curve; d-op = 2D curl
- ∂S = closed curve; d-op = 3D curl
- ∂E = closed surface; d-op = divergence

Dimension of $\partial\Omega$ = dimension of Ω minus 1.

Quick-reference: "What is my differential telling me?"

dt → I'm in 1D parameter land, haven't accounted for curve speed yet

ds → I have true arc length; the speed $|r'|$ is baked in

dr → I have a directed vector; dotting with F gives tangential work per unit length

dA / dx dy → I'm integrating over a flat 2D region

du dv → I'm in 2D parameter land, haven't accounted for surface stretching yet

dS (scalar) → I have true surface area; $|r_u \times r_v|$ is baked in

dS (vector) → I have an oriented surface element; dotting with F gives normal flux

dV → I'm integrating over a 3D region

8 · At-a-Glance Quick Reference

THEOREM	LHS	=	RHS	DIMENSION GAIN	KEY OPERATOR
FTC (line)	$\int_C \nabla f \cdot dr$	=	$f(b)-f(a)$	1D → 0D	gradient ∇f
Green's	$\oint_{\partial D} P \, dx + Q \, dy$	=	$\iint_D (Q_x - P_y) \, dA$	1D → 2D	2D curl (z-component)
Stokes'	$\oint_{\partial S} F \cdot dr$	=	$\iint_S (\text{curl } F) \cdot dS$	1D → 2D	curl $F = \nabla \times F$
Divergence	$\oiint_{\partial E} F \cdot dS$	=	$\iiint_E (\text{div } F) \, dV$	2D → 3D	div $F = \nabla \cdot F$